Factory-to-Customer Shipping Route Efficiency Analysis

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***Abstract*—Efficient logistics management is critical for ensuring timely product delivery and maintaining operational efficiency in supply chain systems. This study presents a comprehensive analysis of factory-to-customer shipping routes using a real-world dataset from a candy distribution network. The objective is to evaluate route efficiency, identify delays, and detect geographic bottlenecks.**

The methodology involves data cleaning, feature engineering, statistical aggregation, and visualization techniques to analyze shipping performance. Key performance indicators such as average lead time, variability, shipment volume, and delay rate are computed to assess operational efficiency.

An interactive dashboard developed using Streamlit enables dynamic exploration of data through filters, maps, and drill-down capabilities. The findings reveal significant variations in route performance, highlight inefficient delivery paths, and demonstrate the impact of shipping modes on delivery speed.

This study emphasizes the importance of data-driven decision-making in logistics and provides a scalable framework for optimizing supply chain operations.

***Index Terms*—Logistics, Supply Chain, Data Analysis, Streamlit, Visualization, Route Optimization**

I. Introduction

In modern supply chain systems, logistics efficiency plays a crucial role in ensuring timely delivery and customer satisfaction. With increasing complexity in distribution networks, organizations face challenges in identifying inefficiencies and optimizing delivery routes. Traditional logistics systems often rely on reactive decision-making, where issues are addressed only after delays occur. This approach leads to increased operational costs and reduced service quality.

This project introduces a data-driven approach to analyze shipping performance across factory-to-customer routes. By leveraging data analytics and visualization techniques, the study provides actionable insights into route efficiency, geographic bottlenecks, and delivery performance. The integration of an interactive dashboard further enhances decision-making by enabling real-time exploration of shipping data.

II. Problem Statement

The organization lacks a systematic approach to evaluate the efficiency of its shipping routes. Specifically, there is limited visibility into:

1) Which routes consistently perform efficiently
2) Which routes experience frequent delays
3) How shipping performance varies across regions and states
4) The impact of different shipping modes on delivery time

Due to this lack of insight, logistics optimization remains reactive rather than proactive. This results in inefficiencies, increased delivery time, and suboptimal resource utilization.

This project aims to address these challenges by developing a data-driven framework for analyzing and visualizing shipping performance.

III. Objectives

The main objectives of this project are:

1) To compute and analyze shipping lead time across routes.
2) To identify efficient and inefficient delivery routes.
3) To detect geographic bottlenecks in logistics operations.
4) To evaluate the performance of different shipping modes.
5) To build an interactive dashboard for data exploration.
6) To enable data-driven decision-making in logistics operations.

IV. System Architecture

To support data-driven logistics optimization

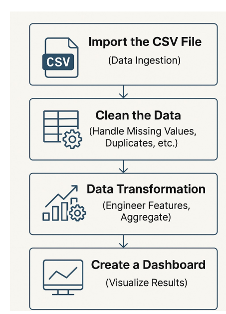

Fig. 1. Data Processing and Visualization Pipeline

The pipeline consists of the following stages:

1) Data Ingestion – Loading raw dataset
2) Data Cleaning – Handling missing and invalid values
3) Feature Engineering – Creating lead time and route features
4) Aggregation – Computing route and state-level metrics
5) Visualization – Generating charts and maps
6) Dashboard – Interactive exploration using Streamlit

V. Datasets Description

The dataset consists of over 10,000 records representing shipping transactions in a candy distribution network. It includes both temporal and geographic attributes, enabling multi-dimensional analysis.

Key features include:

1) Order Date and Ship Date – used to calculate shipping lead time
2) Product Name – used to map products to factories
3) State/Province and Region – used for geographic analysis
4) Ship Mode – used to compare delivery methods
5) Order ID – used for tracking shipment volume

The dataset provides sufficient granularity to analyze performance at route, state, and regional levels.

VI. Methodology

The project follows a structured data analysis approach:

1) **Data Cleaning**: Data cleaning is a critical step to ensure accuracy and reliability. Invalid records with negative lead time were removed, and missing values in key columns were handled. Date columns were converted into date-time format to enable temporal analysis.
2) **Feature Engineering**: Feature engineering enhances the dataset by creating meaningful variables:
   - Shipping Lead Time = Ship Date – Order Date
   - Factory Mapping = Product → Factory
   - Route Creation = Factory → Destination State

   These features enable route-level performance analysis and geographic mapping.
3) **Data Aggregation**: The dataset is aggregated to compute key performance metrics:
   - Average Lead Time → Efficiency indicator
   - Standard Deviation → Variability indicator
   - Volume → Shipment frequency

   These metrics are calculated at both route and state levels.
4) **Delay Classification**: A delay threshold is defined using the **75th percentile** of shipping lead time. Orders exceeding this threshold are classified as delayed, enabling identification of problematic routes.
5) **Visualization**: Multiple visualization techniques are used:
   - Bar charts for route comparison
   - Heatmaps for geographic analysis
   - Scatter plots for bottleneck detection
   - Box plots for distribution analysis
6) **Dashboard Development**: An interactive dashboard was built using Streamlit to allow users to explore insights dynamically.

VII. Data Analysis and Visualization

The analysis revealed significant variations in shipping performance across routes and regions.

1) Route Efficiency Analysis: Route-level analysis reveals significant variation in shipping performance. Efficient routes exhibit low average lead times, while inefficient routes show consistently high delays.

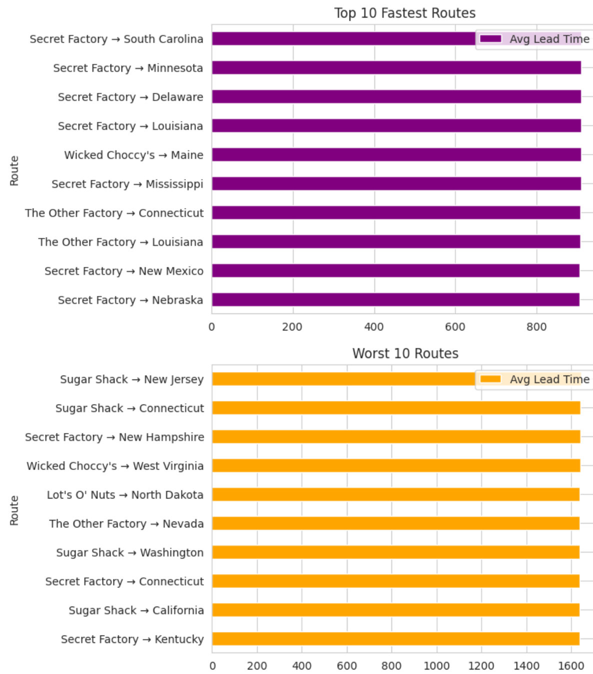

Fig. 2. Route Efficiency Comparison

2) Geographic Bottleneck Analysis: Geographic analysis highlights states with high average lead times, indicating potential bottlenecks. States with both high volume and high lead time represent critical areas requiring immediate attention.

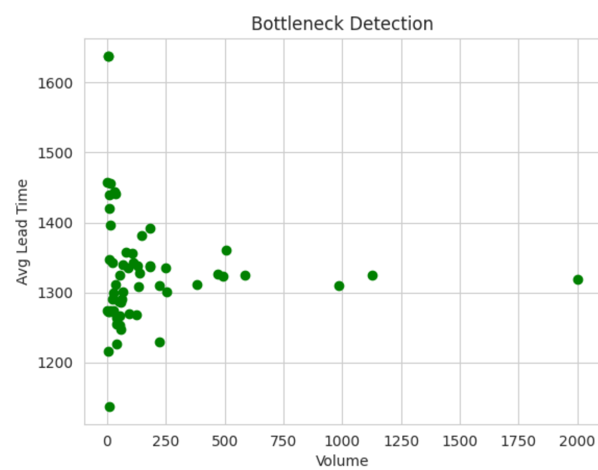

Fig. 3. Bottleneck Analysis

3) Ship mode comparison: Shipping modes significantly influence delivery efficiency. Faster shipping methods consistently show lower lead times compared to standard methods.

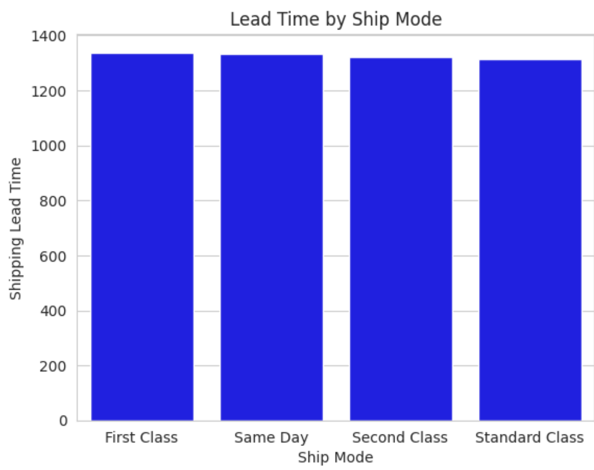

Fig. 4. Ship mode comparison

4) Variability in lead time across routes suggested inconsistency in logistics performance

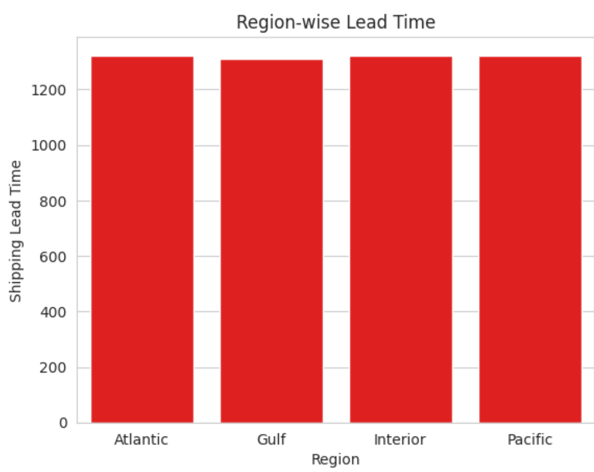

Fig. 5. Lead time Analysis

These findings provide a clear understanding of where improvements are needed in the logistics network.

VIII. Business Insights

The project provides several actionable insights for improving logistics operations:

1) Inefficient routes should be optimized or restructured to reduce delivery time
2) High-delay regions require investigation to identify underlying issues such as infrastructure or demand spikes
3) Shipping methods should be evaluated to prioritize faster and more reliable options
4) Consistency in delivery performance should be improved by reducing variability in lead times
5) Data-driven monitoring systems should be implemented for continuous performance tracking

These insights can help organizations improve delivery efficiency, reduce costs, and enhance customer satisfaction.

IX. Dashboard Streamlit

An interactive dashboard was developed using Streamlit to visualize the analysis results. The dashboard includes multiple modules such as route efficiency overview, geographic shipping map, ship mode comparison, and route drill-down.

Users can apply filters based on date range, region, state, and shipping mode to explore data dynamically. The dashboard also provides key performance indicators such as average lead time, total orders, and delay rate.

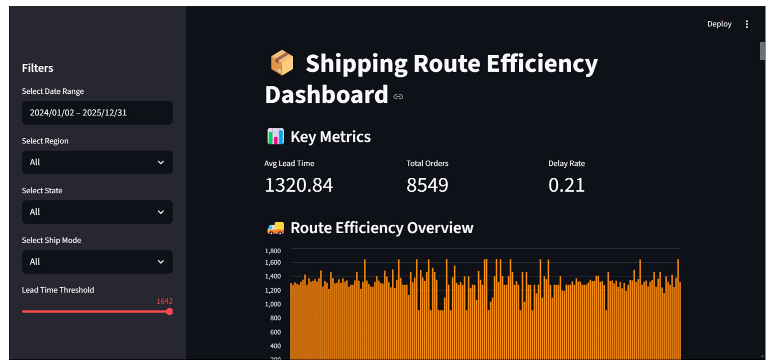

Fig. 6. Dashboard Streamlit

X. Future Scope

Future enhancements for this project may include:

1) Integration of real-time data for live monitoring
2) Implementation of machine learning models to predict delays
3) Optimization algorithms for route planning
4) Inclusion of cost analysis for better decision-making
5) Deployment of the dashboard as a web application for wider accessibility

These improvements can further enhance the system's capability and provide deeper insights into logistics performance.

XI. Conclusion

This project successfully demonstrates how data analysis and visualization can be used to evaluate and improve logistics performance. By analyzing shipping data and developing an interactive dashboard, the study provides valuable insights into route efficiency, delays, and bottlenecks.

The results highlight the importance of data-driven decision-making in supply chain management. The developed system enables proactive identification of issues, allowing organizations to optimize their logistics operations effectively.